

Neural Network Quantum States enhanced by the Lanczos step

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Abstract

Accurate simulation of strongly correlated N-body quantum systems remains a fundamental challenge, particularly in frustrated regimes where the sign problem restricts exact quantum Monte Carlo (QMC) methods. In this work, we use the variational Monte Carlo (VMC) method, combined with quantum states of neural networks (NQS), to investigate the ground-state properties of quantum spin networks. In order to systematically improve the basic variational method of the restricted Boltzmann machine (RBM) and mitigate its intrinsic biases, we apply a one-step Lanczos projection. We use two complementary techniques: an optimisation based on the variational Lanczos parameter, and an approach based on exact orthogonalisation of the Krylov subspace to estimate the Lanczos energy.

1 Introduction

Accurately simulating strongly-correlated quantum many-body systems is notoriously difficult, particularly in frustrated regimes where the sign problem restricts the use of exact QMC methods [1, 2, 3]. Although traditional methods are very efficient, their extension to two-dimensional or frustrated networks often remains prohibitively computationally expensive. Consequently, VMC combined with NQS has emerged as a scalable and powerful alternative [4].

In this work, we study the isotropic spin-1/2 antiferromagnetic $J_1 - J_2$ Heisenberg model, which introduces competing interactions and geometrical frustration. The general Hamiltonian is given by:

$$\mathcal{H} = J_1 \sum_{\langle i,j \rangle} \mathbf{S}_i \cdot \mathbf{S}_j + J_2 \sum_{\langle\langle i,j \rangle\rangle} \mathbf{S}_i \cdot \mathbf{S}_j \quad (1)$$

where $\mathbf{S}_i = (S_i^x, S_i^y, S_i^z)$ is the quantum spin-1/2 operator at site i . The parameters J_1 and J_2 (with $J_1, J_2 > 0$, $J_1 = 1.0$) denote the nearest-neighbor $\langle i, j \rangle$ and next-nearest-neighbor $\langle\langle i, j \rangle\rangle$ antiferromagnetic exchange couplings, respectively. The standard unfrustrated Heisenberg model is naturally recovered by setting $J_2 = 0$.

While standard VMC provides a strong variational baseline, the resulting wavefunctions can be systematically improved by applying a few Lanczos

steps [5, 6, 7]. Evaluating both the expected energy and its variance after the Lanczos step enables a zero-variance extrapolation. This technique makes it possible to obtain estimates of the ground state energy, thus effectively mitigating the intrinsic biases of the initial ansatz of the neural network.

All numerical simulations presented in this study are implemented using the **NetKet** machine learning framework [8]. The complete source code developed for this work are openly available at: <https://github.com/victor-pc11/Lanczos-Algorithm-Ground-State-Heisenberg-model/tree/main>.

2 Theory

We investigate two distinct numerical methodologies for evaluating this Lanczos state. Throughout this study, we restrict our comparative analysis to the single-step projection ($p = 1$), formally defining the improved trial state as $|\Psi_L\rangle \equiv |\Psi_{p=1}\rangle$. The three geometric configurations shown in the Figure 1 constitute standard and widely studied reference models. Their finite dimensions allow the calculation of the exact ground state energy per site via exact diagonalization (ED), thus providing a reference for our variational methods. The exact reference energies E_0^{exact} are: -0.4452 for the 1D chain ($L = 20$); -0.7018 for the 2D square (4×4); and -0.5286 for the frustrated model $J_1 - J_2$ (4×4 , $J_2/J_1 = 0.5$).

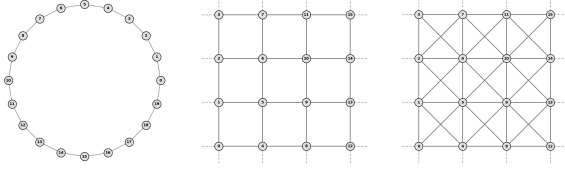


Figure 1: Visual representation of the studied lattice geometries. (Left) Standard 1D Heisenberg chain ($L = 20$). (Center) A 4×4 square lattice with periodic boundary conditions (PBC) and strictly nearest-neighbor interactions. (Right) The frustrated $J_1 - J_2$ model on a 4×4 lattice with PBC, where next-nearest-neighbor (NNN) couplings introduce competing magnetic interactions of order J_2 .

2.1 Variational Wave Functions

To approximate the ground state of the Hamiltonian, we employ VMC with a NQS ansatz. Specifically, our variational wave function $|\Psi_v\rangle$ is parametrized by a RBM with a hidden-unit feature density of $\alpha = 1$, equating the number of hidden neurons to the number of physical lattice sites.

Energy expectation values and parameter gradients are evaluated stochastically via Markov Chain Monte Carlo (MCMC) sampling. We utilize a Metropolis-Exchange sampler drawing $N_{\text{samples}} = 2048$ samples per iteration. By proposing updates that exclusively exchange the states of neighboring spins, this sampler strictly conserves the total system magnetization ($S_z = 0$), reducing the size of the explored Hilbert space.

The RBM parameters are optimized using Stochastic Reconfiguration (SR). Unlike standard gradient descent, SR acts as a quantum natural gradient method that accounts for the geometric structure (the Fubini-Study metric) of the variational manifold. To ensure the numerical stability of the covariance matrix inversion during the SR step, a diagonal shift regularization of $\epsilon = 10^{-4}$ is applied. Parameter updates are driven by Stochastic Gradient Descent (SGD) with a learning rate of $\eta = 0.01$. After a few iterations, the network converges to a stable minimum on the RBM manifold. This optimized state, $|\Psi_v\rangle$, captures the essential properties of the ground state and constitutes the starting point for the subsequent Lanczos step.

2.2 Variational Optimization of the Lanczos Parameter

To evaluate the energy of the first Lanczos step, we optimize the variational parameter α [6]. To ensure robust convergence of this parameter, we first reconstruct the continuous energy landscape of the projected state $|\Psi_L\rangle$ using a stochastic reweighting technique, then locate the exact minimum using gradient-based optimization.

2.2.1 Stochastic Reweighting and Energy Landscape

Rather than running a computationally expensive new MCMC simulation for every trial value of α , we evaluate the expected energy of the Lanczos state $E_L(\alpha)$ as a weighted statistical average over the baseline RBM configurations $\{x_i\}$:

$$E_L(\alpha) \approx \frac{\sum_{i=1}^N W(x_i, \alpha) E_{\text{loc}, L}(x_i)}{\sum_{i=1}^N W(x_i, \alpha)}. \quad (2)$$

Here, the reweighting factor $W(x, \alpha) = |1 + \alpha E_{\text{loc}, v}(x)|^2$ accounts for the change in the probability measure. By applying the linearity of the Hamiltonian, the local energy of the Lanczos state is derived exclusively from the initial local estimators:

$$E_{\text{loc}, L}(x) = \frac{E_{\text{loc}, v}(x) + \alpha H_{\text{loc}, v}^2(x)}{1 + \alpha E_{\text{loc}, v}(x)}. \quad (3)$$

This strategy rapidly reconstructs the continuous energy landscape $E_L(\alpha)$ on a dense grid of α values using pre-computed local moments. However, the initial calculation of $H_{\text{loc}, v}^2(x)$ remains a bottleneck in the Lanczos projection.

2.2.2 Stochastic Gradient Descent Optimization

We fix all internal parameters of the convergent state $|\Psi_v\rangle$ and minimise the energy using the stochastic gradient descent (SGD) method. In order to overcome the topological difficulties posed by the energy landscape resulting from reweighting, optimization of α requires careful initialization. The parameter is initialized to the right of the unstable region at $\alpha_{\text{init}} = 0.5$, with a learning rate of $\eta = 10^{-2}$ over 300 iterations. These hyperparameters allow convergence to the value of α estimated by the reweighting method in the geometries studied without frustration. Furthermore, α is initialized as a complex number for the frustrated geometry.

2.3 Exact Diagonalization in the Orthogonalized Krylov Subspace

This method calculates the exact energy of the Lanczos state by solving the eigenvalue problem in a 2-dimensional Krylov subspace. Instead of using the naive basis $\{|\psi_0\rangle, H|\psi_0\rangle\}$, which leads to the evaluation of raw moments ($\langle H^2 \rangle$, $\langle H^3 \rangle$) and suffers heavily from catastrophic cancellation [7], we follow the approach detailed in Appendix D of Roth *et al.* (2023) [9]. We define an orthogonalized and normalized basis component:

$$|\psi_1\rangle = \frac{H - \langle H \rangle}{\sigma} |\psi_0\rangle \quad (4)$$

where $\sigma^2 = \langle H^2 \rangle - \langle H \rangle^2$ is the variance. In this refined basis, the 2×2 Hamiltonian matrix H_L takes the form:

$$H_L = \langle H \rangle + \sigma \begin{pmatrix} 0 & 1 \\ 1 & \mu_3 \end{pmatrix} \quad (5)$$

where μ_3 is the standardized third moment (skewness). The exact Lanczos energy is simply the lowest eigenvalue of this matrix. Given $E_0 = \langle H \rangle$, the first two shifted estimators for a given spin configuration x are:

$$\begin{aligned} \varepsilon_{loc}(x) &= E_{loc}(x) - E_0 \\ \nu_{loc}(x) &= E_{loc}^{H^2}(x) - 2E_0 E_{loc}(x) + E_0^2 \end{aligned} \quad (6)$$

From these, the matrix parameters are reliably extracted via standard statistical functions over the Monte Carlo samples: variance: $\sigma^2 = \langle \nu_{loc} \rangle$ and skewness: $\mu_3 = \frac{\text{Cov}(\varepsilon_{loc}, \nu_{loc})}{\sigma^3}$. The lowest eigenvalue of H_L , which corresponds to the exact Lanczos energy, is then given analytically by:

$$E_L = \langle H \rangle + \sigma \left(\frac{\mu_3}{2} - \sqrt{\frac{\mu_3^2}{4} + 1} \right) \quad (7)$$

This method therefore allows us to estimate the Lanczos energy by calculating only the moments $\langle H \rangle$ and $\langle H^2 \rangle$.

2.4 Zero-Variance Extrapolation

In addition to the direct evaluation of Lanczos energy E_L , the energy of the ground state E_0 can be estimated by zero-variance extrapolation. For a sufficiently precise test state $|\Psi_p\rangle$ and for a small variance, its expected energy per site, $E_p = \langle H \rangle / N$, is linearly proportional to the associated energy variance per site, $\sigma_p^2 = (\langle H^2 \rangle - \langle H \rangle^2) / N$:

$$E_p \approx E_0 + c \sigma_p^2. \quad (8)$$

By plotting E_p against σ_p^2 at different stages of optimization and performing a linear regression, E_0 is reliably extracted at the zero-variance intercept ($\sigma_p^2 = 0$).

This linear scaling strictly assumes that the trial state is already a highly accurate approximation of the true ground state. In this asymptotic regime, if the state deviates by an orthogonal perturbation of order $\mathcal{O}(\epsilon)$, both the energy error ($E_p - E_0$) and the variance σ_p^2 scale as $\mathcal{O}(\epsilon^2)$, ensuring a strict linear dependence. Conversely, for poorly optimized states characterized by large variances, higher-order terms become non-negligible, and the extrapolation becomes unreliable. To quantify the uncertainty of this extrapolation, we employ a parametric bootstrap method. We generate virtual Gaussian samples for each data point based on its Standard Error of the Mean (SEM). By extracting the y-intercepts of independent linear fits across these virtual datasets, we construct a probability distribution for the ground-state energy.

3 Results

3.1 Variational Optimization of the Lanczos Parameter

Table 1 compares the baseline Variational Monte Carlo energy (E_{VMC}) and the single-step Lanczos energy (E_L) for the 1D Heisenberg chain ($L = 20$) and the 2D unfrustrated square lattice (4×4). To

System	it _{init}	E_{VMC}	E_L
Chain	50	-0.4357 ± 0.0034	-0.4431 ± 0.0020
	75	-0.4420 ± 0.0019	-0.4446 ± 0.0012
	100	-0.4440 ± 0.0011	-0.4449 ± 0.0008
Square	100	-0.6982 ± 0.0140	-0.7014 ± 0.0039
	150	-0.6980 ± 0.0147	-0.7014 ± 0.0039
	200	-0.6984 ± 0.0139	-0.7014 ± 0.0041

Table 1: Ground-state energies per site for variational optimization of the Lanczos parameter. Compares the baseline VMC energy (E_{VMC}) and the single-step Lanczos energy (E_L) at various optimization stages.

highlight the corrective power of the variational optimization method, we evaluated these energies at three intermediate stages of the initial VMC optimization (it_{init}). These checkpoints represent states that are reasonably converged yet sufficiently distant from the true ground state, isolating the exact energy gain provided by the Lanczos projection.

The highly frustrated 2D $J_1 - J_2$ model is excluded from this comparison, as its explicit Lanczos energy computation proved computationally prohibitive without cluster resources.

Using the zero-variance extrapolation procedure detailed in Sect. 2.4, we obtain the extrapolated ground-state energies shown in Figure 2.

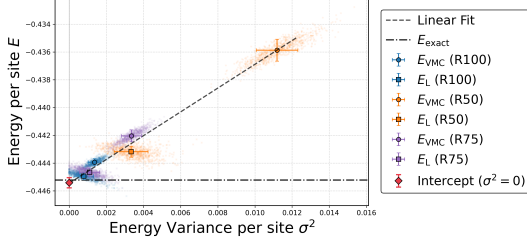


Figure 2: Zero-variance extrapolation for a 1D chain ($L = 20$) across multiple optimisation runs (R). Baseline VMC states (circles) and Lanczos projections (squares) are shown for various check-points (colors). Transparent clouds indicate raw MCMC samples, while solid markers denote means with Standard Errors. The linear fit (black dashed line) intercepts the zero-variance axis (red diamond), matching the exact ground state (E_{exact}).

3.2 Exact Diagonalization in the Orthogonalized Krylov Subspace

Table 2 presents the Lanczos step energies obtained via exact subspace diagonalization after an initial VMC optimization phase, as described in Sect. 2.3. Applying the zero-variance extrapolation procedure

System	it _{VMC}	E_{VMC}	E_L
Chain	50	-0.4359 ± 0.0033	-0.4432 ± 0.0018
	75	-0.4420 ± 0.0018	-0.4447 ± 0.0011
	100	-0.4439 ± 0.0012	-0.4449 ± 0.0009
Square	100	-0.6980 ± 0.0044	-0.7015 ± 0.0020
	150	-0.6985 ± 0.0043	-0.7016 ± 0.0018
	200	-0.6985 ± 0.0043	-0.7017 ± 0.0020

Table 2: Ground-state energies per site for exact diagonalization in the orthogonalized Krylov subspace. Compares the baseline VMC energy (E_{VMC}) and the single-step Lanczos energy (E_L) at various optimization stages.

detailed in Sect. 2.4 to these data points, we construct a probability distribution for the extrapolated intercept. The results of the extrapolation for non-frustrated geometries are summarised in the Table 3.

3.3 Energy Landscape and Variational Optimization

Using the stochastic reweighting technique (Sect. 2.2), we evaluate the energy landscape $E_L(\alpha)$. As shown in Figure 3, a topological feature emerges: a sharp energy barrier slightly above $\alpha = 0$ separates a monotonic left tail from the true variational minimum. This feature is present in all the geometries examined in this study

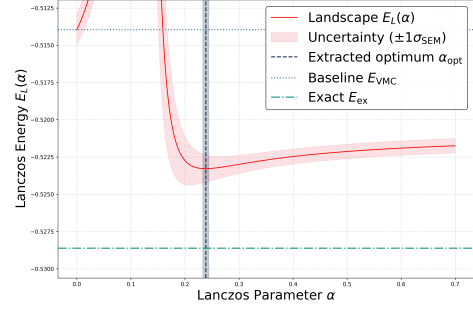


Figure 3: Reconstructed energy landscape $E_L(\alpha)$ computed via stochastic reweighting for a 2D frustrated model (4×4 , $J_2/J_1 = 0.5$).

To determine the optimal parameter α_{opt} within this noisy landscape, we implement a bootstrap algorithm [10]. We generate virtual landscapes via parametric Gaussian resampling based on the standard error of the mean (SEM) obtained by the VMC method. Extracting the absolute minimum from these virtual curves yields a probability distribution for α_{opt} .

Informed by this landscape analysis, our explicit SGD optimization (initialized safely at $\alpha_{\text{init}} = 0.5$) successfully bypasses the energy barrier. It descends towards the corresponding well and oscillates around the optimal value, as illustrated in Figure 4 for a chain where $\alpha_{\text{opt}} = 0.205 \pm 0.005$.

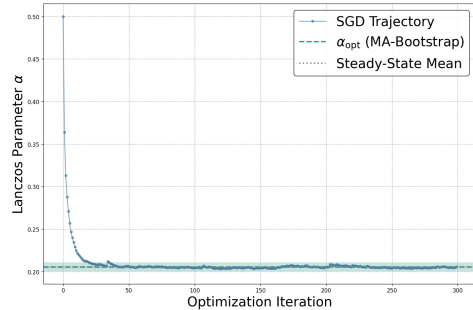


Figure 4: SGD trajectory of the Lanczos parameter α for a 1D chain ($L = 20$). Starting from $\alpha_{\text{init}} = 0.5$, the parameter converges to the estimated minimum.

We further test this framework under strong geometric frustration. Applying the Lanczos iteration to the $J_1 - J_2$ model similarly yields a rapid descent and a stable convergence plateau for α (Figure 5). Interestingly, we observe that the SGD trajectory converges to a plateau distinct from the minimum predicted by the reweighting procedure, highlighting the limitations of the reweighting approximation in highly frustrated regimes.

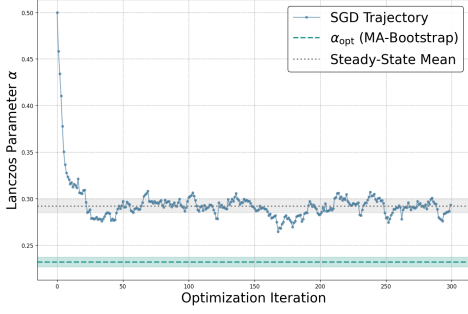


Figure 5: SGD trajectory of the real part of the Lanczos parameter α for the strongly frustrated $J_1 - J_2$ lattice ($\alpha_{\text{init}} = 0.5$, $J_2/J_1 = 0.5$). The imaginary part oscillates stably around zero.

Despite this discrepancy in the reweighting estimate, the dual optimization strategy still delivers a clear sequential improvement. As shown in Figure 6, applying a single iteration of the Lanczos algorithm minimises the local energy and significantly reduces the variance compared with the raw VMC baseline, confirming the corrective power of the projection.

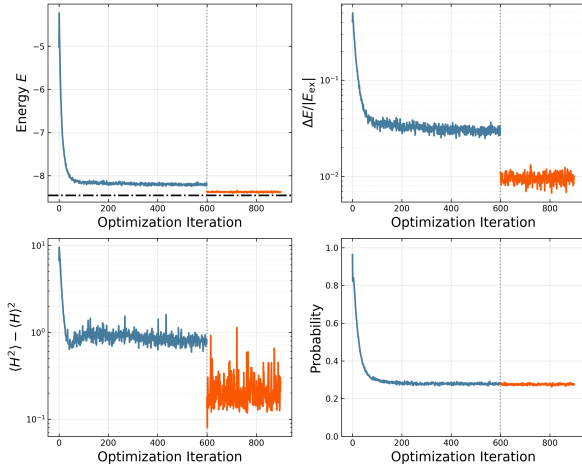


Figure 6: Quantum state optimization metrics in a frustrated lattice (4×4 , $J_2/J_1 = 0.5$). The initial VMC phase is shown in blue, followed by the SGD optimization of the Lanczos parameter α , shown in orange. The exact ground-state energy E_{exact} is marked by the black dash-dotted line.

3.4 Summary of Ground-State Energy Estimates

To conclude our comparative analysis, Table 3 summarizes the final ground-state energy estimates obtained for each lattice geometry. We compare the exact reference energies (E_{exact}) with the zero-variance extrapolations obtained via our two single-step Lanczos methods (Sect. 2.4).

System	E_{exact}	$E_0^{\text{v.o.}}$	$E_0^{\text{e.d.}}$
Chain	-0.4452	-0.4453 ± 0.0004	-0.4454 ± 0.0004
Square	-0.7018	-0.7015 ± 0.0003	-0.7019 ± 0.0009

Table 3: Extrapolated ground-state energies. The exact reference (E_{exact}) is compared to zero-variance estimates from Variational Optimization ($E_0^{\text{v.o.}}$) and Krylov Exact Diagonalization ($E_0^{\text{e.d.}}$) for the 1D chain ($L = 20$) and 2D square lattice (4×4 PBC).

Ultimately, both Lanczos projection techniques demonstrate substantial improvements in the ground-state estimates compared to the raw VMC baseline. While exact diagonalization in the Krylov subspace ($E_0^{\text{e.d.}}$) proves computationally more straightforward, both methods yield highly consistent results for unfrustrated geometries, achieving comparable accuracy in the extrapolated energies (Table 3).

4 Future Work

In order to improve the variational optimisation approach for frustrated geometries, future work should focus on refining the initial reference state $|\Psi_v\rangle$ before applying the Lanczos step, or on exploring the multi-step Lanczos method.

Furthermore, future research could aim to extend the study to new complex and highly frustrated topological geometries. Promising candidates include Kagome-type and triangular spin lattices, where the severity of the sign problem severely limits classical exact numerical methods.

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References

- [1] G. Carleo. *Advanced computational physics*.
- [2] G. Carleo. *Computational Quantum Physics*.
- [3] D. Wu, R. Rossi, F. Vicentini, N. Astrakhantsev, F. Becca, X. Cao, J. Carrasquilla, F. Ferrari, A. Georges, M. Hibat-Allah, M. Imada, A. M. Läuchli, G. Mazzola, A. Mezzacapo, A. Millis, J. Robledo Moreno, T. Neupert, Y. Nomura, J. Nys, O. Parcollet, R. Pohle, I. Romero, M. Schmid, J. M. Silvester, S. Sorella, L. F. Tocchio, L. Wang, S. R. White, A. Wietek, Q. Yang, Y. Yang, S. Zhang, and G. Carleo. “Variational benchmarks for quantum many-body problems”. In: *Science* 386.6719 (Oct. 18, 2024), pp. 296–301. DOI: [10.1126/science.adg9774](https://doi.org/10.1126/science.adg9774). URL: <https://www.science.org/doi/10.1126/science.adg9774> (visited on 05/11/2026).
- [4] G. Carleo and M. Troyer. “Solving the Quantum Many-Body Problem with Artificial Neural Networks”. In: *arXiv.org* (June 7, 2016). DOI: [10.1126/science.aag2302](https://doi.org/10.1126/science.aag2302). URL: <https://arxiv.org/abs/1606.02318v1> (visited on 05/08/2026).
- [5] K. Loehr and B. K. Clark. *Enhancing Neural Network Backflow*. Oct. 30, 2025. DOI: [10.48550/arXiv.2510.26906](https://doi.org/10.48550/arXiv.2510.26906). arXiv: [2510.26906\[cond-mat\]](https://arxiv.org/abs/2510.26906). URL: <http://arxiv.org/abs/2510.26906> (visited on 05/08/2026).
- [6] F. Becca, W.-J. Hu, Y. Iqbal, A. Parola, D. Poilblanc, and S. Sorella. “Lanczos steps to improve variational wave functions”. In: *Journal of Physics: Conference Series* 640 (Sept. 28, 2015), p. 012039. DOI: [10.1088/1742-6596/640/1/012039](https://doi.org/10.1088/1742-6596/640/1/012039). arXiv: [1412.2656\[cond-mat\]](https://arxiv.org/abs/1412.2656). URL: <http://arxiv.org/abs/1412.2656> (visited on 05/08/2026).
- [7] E. R. Gagliano, E. Dagotto, A. Moreo, and F. C. Alcaraz. “Correlation functions of the antiferromagnetic Heisenberg model using a modified Lanczos method”. In: *Physical Review B* 34.3 (Aug. 1, 1986), pp. 1677–1682. DOI: [10.1103/PhysRevB.34.1677](https://doi.org/10.1103/PhysRevB.34.1677). URL: <https://link.aps.org/doi/10.1103/PhysRevB.34.1677> (visited on 05/23/2026).
- [8] *NetKet: The Machine Learning Toolbox for Quantum Physics — NetKet*. URL: <https://netket.readthedocs.io/en/latest/index.html> (visited on 05/23/2026).
- [9] C. Roth, A. Szabó, and A. H. MacDonald. “High-accuracy variational Monte Carlo for frustrated magnets with deep neural networks”. In: *Physical Review B* 108.5 (Aug. 8, 2023), p. 054410. DOI: [10.1103/PhysRevB.108.054410](https://doi.org/10.1103/PhysRevB.108.054410). URL: <https://link.aps.org/doi/10.1103/PhysRevB.108.054410> (visited on 05/08/2026).
- [10] G. Cowan and G. Cowan. *Statistical Data Analysis*. Oxford, New York: Oxford University Press, Mar. 26, 1998. 212 pp.